Watch. Listen. Read.

Spotting hate in online videos by combining sight, sound, and speech

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Problem definition

Problem. Hateful content online increasingly travels as video, where meaning is carried jointly by visual context, audio delivery, and the spoken message. Single-modality classifiers struggle on cases where these three signals must be combined to distinguish hate from satire or critical discourse.

Hypothesis. A small transformer trained on top of frozen pretrained per-modality encoders can outperform every unimodal baseline on a real corpus of hateful and non-hateful videos, with gains that are not uniform across target communities.

Dataset

HateMM (Das et al., ICWSM 2023), released under CC BY 4.0 on Zenodo. We use it as-is, without modification or redistribution.

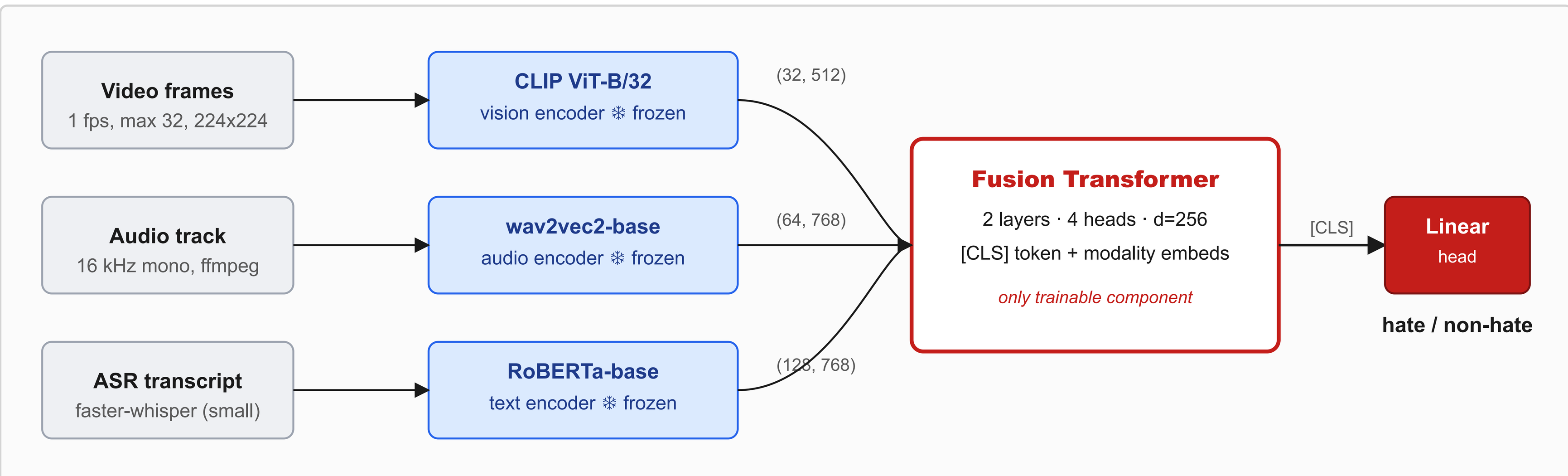
- **1083 BitChute videos** (431 hate, 652 non-hate); ~43.5 hours, 6.3 GB.
- A video is labeled *hate* when it attacks or demeans someone for who they are: religion, race, ethnicity, gender, sexual orientation, or disability. Each hate video also tags the targeted community.
- 80/10/10 stratified split by binary label (863/109/109), fixed seed; preprocessing succeeded on 1081/1083 videos.
- **Scope:** BitChute is a low-moderation platform, so our results characterise hate detection on that source distribution; transfer to YouTube or TikTok would require recalibration.

Method

Pipeline in three stages.

1. **Preprocess.** Sample frames at 1 fps capped at 32 per video, center-cropped to 224 x 224; extract 16 kHz mono audio via ffmpeg; transcribe speech with faster-whisper.
2. **Frozen feature extraction.** CLIP ViT-B/32 per frame, wav2vec2-base mean-pooled into 64 audio tokens, RoBERTa-base token embeddings (length 128) on the transcript.
3. **Fusion.** A 2-layer transformer (d = 256, 4 heads, GELU, pre-norm) ingests the concatenated per-modality token sequences plus learnable modality-type embeddings and a [CLS] token; the [CLS] output goes through a linear head for binary classification.

Why frozen encoders. Keeping all three encoders frozen makes the modality ablation a fair comparison: only the small fusion head varies between configurations. Restricting the same architecture to a single modality at the input gives our unimodal baselines.



Related works

- **HateMM** [1]. Defines the task and releases the dataset and target-community labels we audit against; their best multimodal baseline reaches ~0.80 macro-F1.
- **Hateful Memes** [2]. Foundational multimodal hate benchmark (image + text); MOMENTA [3] adds context-aware fusion on the same line of work.
- **Attention Bottlenecks** (Nagrani et al.) [4]. Motivate cross-modal attention; we adopt a lighter [CLS]-readout variant for our scale.
- **Frozen encoders:** CLIP [5] for vision, wav2vec2 [6] for audio, RoBERTa [7] for text.

Plot twist

Our **first run, with one random seed**, showed fusion winning by 5 accuracy points and we almost wrote a paper claiming «fusion wins everywhere.» Re-running across three seeds {42, 1337, 2025} made that gap **vanish into seed variance**; only the AUROC gain held up.

Lesson. A single seed can deceive. Multi-seed analysis is what separates a confident headline from an honest one.

References

1. M. Das et al., "HateMM: A Multi-Modal Dataset for Hate Video Classification," *Proc. ICWSM*, 2023.

5. A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision (CLIP)," *Proc. ICML*, 2021.

Validation

Table 1. Modality ablation on the test split (n = 109), mean ± std over 3 seeds {42, 1337, 2025}. Our 0.78 macro-F1 is comparable to Das et al.'s reported ~0.80 on the same dataset.

Modality	Acc	F1	Prec	Rec	AUROC
V + A + T	**0.79 ± 0.05**	**0.78 ± 0.05**	0.74 ± 0.08	**0.74 ± 0.08**	**0.88 ± 0.01**
V + A	0.77 ± 0.01	0.76 ± 0.00	0.71 ± 0.04	0.71 ± 0.08	0.85 ± 0.01
V + T	0.78 ± 0.03	0.77 ± 0.02	0.73 ± 0.07	0.72 ± 0.04	0.86 ± 0.01
A + T	0.76 ± 0.02	0.75 ± 0.02	0.71 ± 0.05	0.69 ± 0.08	0.84 ± 0.00
Text only	**0.79 ± 0.02**	**0.78 ± 0.02**	**0.75 ± 0.06**	0.73 ± 0.08	0.83 ± 0.02
Video only	0.72 ± 0.03	0.71 ± 0.02	0.64 ± 0.06	0.68 ± 0.06	0.81 ± 0.01
Audio only	0.67 ± 0.02	0.67 ± 0.03	0.56 ± 0.02	0.71 ± 0.05	0.77 ± 0.03

Table 2. Per-target-community test accuracy (n ≥ 15), mean ± std over 3 seeds.

Community	n	Fusion	Video	Audio	Text
Blacks	46	**0.75 ± 0.05**	0.70 ± 0.04	0.59 ± 0.06	0.73 ± 0.03
Others	39	0.84 ± 0.01	0.74 ± 0.09	0.68 ± 0.11	**0.85 ± 0.05**
Jews	15	0.73 ± 0.13	0.67 ± 0.07	0.71 ± 0.04	**0.76 ± 0.04**

⚠ **Others (n = 39)**: this subset is predominantly non-hate videos, so the model scores high accuracy simply by predicting the majority class — not because it genuinely discriminates hate content in that group.

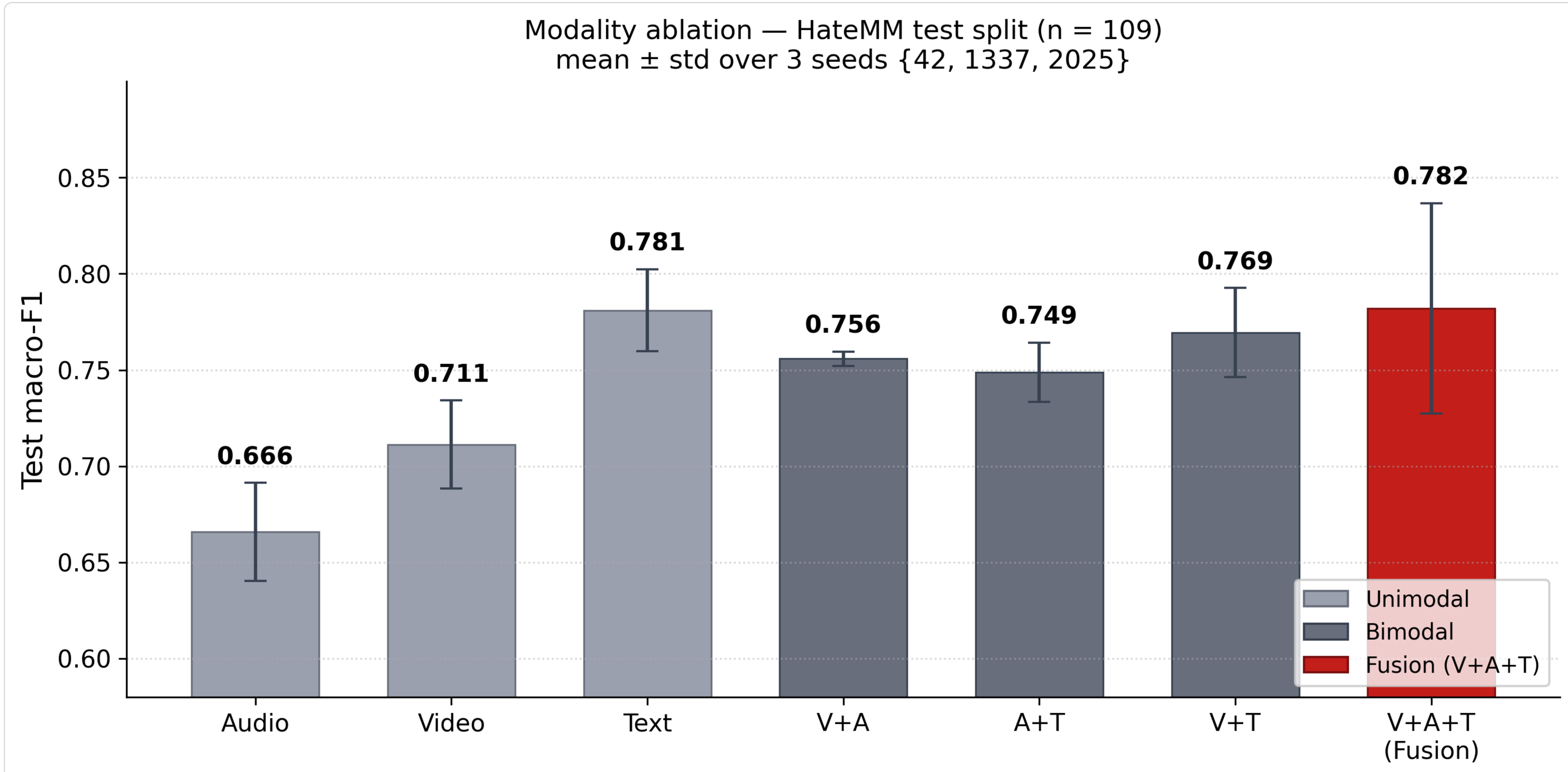


Figure 1. Test macro-F1 for all seven modality configurations, mean ± std over 3 seeds {42, 1337, 2025}.

Key observations

Each added modality lifts AUROC monotonically. From audio alone (0.77) to full fusion (0.88), every step in the seven-row ablation buys ranking signal. Accuracy, in contrast, plateaus at the text-only level (0.79).

Video adds more than audio on top of text. Going text → V+T adds +0.030 AUROC; text → A+T adds only +0.012. V+T and V+A themselves are tied within seed variance. The third modality on top of any pair adds another ~+0.02.

Per-community, text wins *Others* (0.86), V+A+T ties V+T on *Blacks* (0.75), and *Jews* (n = 15) is too noisy to call. No single recipe wins everywhere.

Limitations

- **Small test set (n = 109).** Per-community stats for groups with n < 15 are excluded; reported groups still carry appreciable variance.
- **Heuristic community-label normalisation.** Multi-target labels canonicalised to a sorted comma-joined form, conflating intersectional rows with singleton ones.
- **ASR quality varies.** Shouting, music, and accented speech degrade the text modality on a subset.
- **Architectural choices.** Frozen backbones cap absolute performance, and wav2vec2 hidden states are pooled to a fixed 64 tokens (over-averaging long videos).

Conclusion

Fusing three frozen pretrained encoders through a small transformer head **robustly improves AUROC** over every unimodal baseline on HateMM (0.88 ± 0.01 vs 0.83 ± 0.02 for text). On threshold-dependent metrics the gain washes out into seed variance, a finding single-seed reporting would have missed. Per-community, text alone is competitive with or better than fusion on every reported group, so fusion delivers a calibrated ranking gain rather than a uniform threshold-level improvement over a strong text-only baseline.

Practical implication. Three deployment options on this dataset, choose by your cost/quality trade-off: **(i) V+A+T (AUROC 0.88 ± 0.01)** when ranking quality matters most; **(ii) V+T (AUROC 0.86 ± 0.02)** as a cost-aware sweet spot — within 0.02 AUROC of full fusion with no audio pipeline at all, capturing ~60% of the AUROC lift over text-only; **(iii) Text-only (acc 0.79 ± 0.02)** for fixed-threshold binary decisions — cheapest pipeline and ties V+A+T on accuracy.